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COURSE :CSA0311 Data Structures
Assessment: CO5 - AT2

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CO5-AT2-Short Answer Test

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QUESTIONS

Q1

What is another term for a vertex?
2 marks

✓

Q2

What type of graph has weighted edges?
12 marks

✓

Q3

Does a Minimum Spanning Tree contain cycles?
2 marks

✓

Q4

What does a Minimum Spanning Tree minimize?
1 mark

✓

Q5

Which algorithm is commonly used in GPS navigation systems?
1 mark

✓

5/5 answered

Q1

What is another term for a vertex?
2 marks

What is another term for a vertex?

Your Answer

Another term for a vertex in graph theory is a node. Vertices and nodes are used interchangeably to represent the fundamental units or points where edges connect.

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1 mark

✓

5/5 answered

Q2

What type of graph has weighted edges?
12 marks

What type of graph has weighted edges?

Your Answer

A weighted graph (or a network) is a graph in which every edge is assigned a specific numerical value, weight, or cost.

Definition: Unlike unweighted graphs where edges only indicate connections, a weighted graph incorporates quantitative values on its edges $e = (u, v)$ to represent real-world metrics like physical distance, travel time, transmission costs, or financial expense.
Applications: They are fundamental in modeling complex networks, such as transportation systems, computer networks, and social connections, where efficiency or cost optimization is required.

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1 mark

Q5
Which algorithm is commonly used in GPS navigation systems?
1 mark

5/5 answered

Q3 Does a Minimum Spanning Tree contain cycles?
2 marks

Does a Minimum Spanning Tree contain cycles?

Your Answer
No, a Minimum Spanning Tree (MST) does not contain cycles. By definition, a tree is a connected graph containing no circuits or loops. If a cycle were to be introduced, it would no longer qualify as a tree structure.

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1 mark

5/5 answered

Q4 What does a Minimum Spanning Tree minimize?
1 mark

What does a Minimum Spanning Tree minimize?

Your Answer
A Minimum Spanning Tree minimizes the total edge weight (or total cost) of a connected, undirected graph. It ensures that all vertices are included with the minimum possible sum of edge costs while forming a tree structure with no cycles.

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Which algorithm is commonly used in GPS navigation systems?
1 mark

5/5 answered

Q5 Which algorithm is commonly used in GPS navigation systems?
1 mark

Which algorithm is commonly used in GPS navigation systems?

Your Answer
Dijkstra's algorithm is commonly used in GPS navigation systems because it efficiently computes the shortest path between a starting point and a destination in a weighted graph.

Save Answer

